

LLAM LRI 2026

GenAI Overview and Problems



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- ❖ Manage People's Law Library website
 - ❖ [Member, Maryland Judiciary's AI Governance Subcommittee](#)
 - ❖ Co-Chair, MSBA AI & Legal Technology Task Force
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Topics

- What is AI and how does it work?
- Where AI goes wrong in law practice
- Epistemology, semantics, and the human element

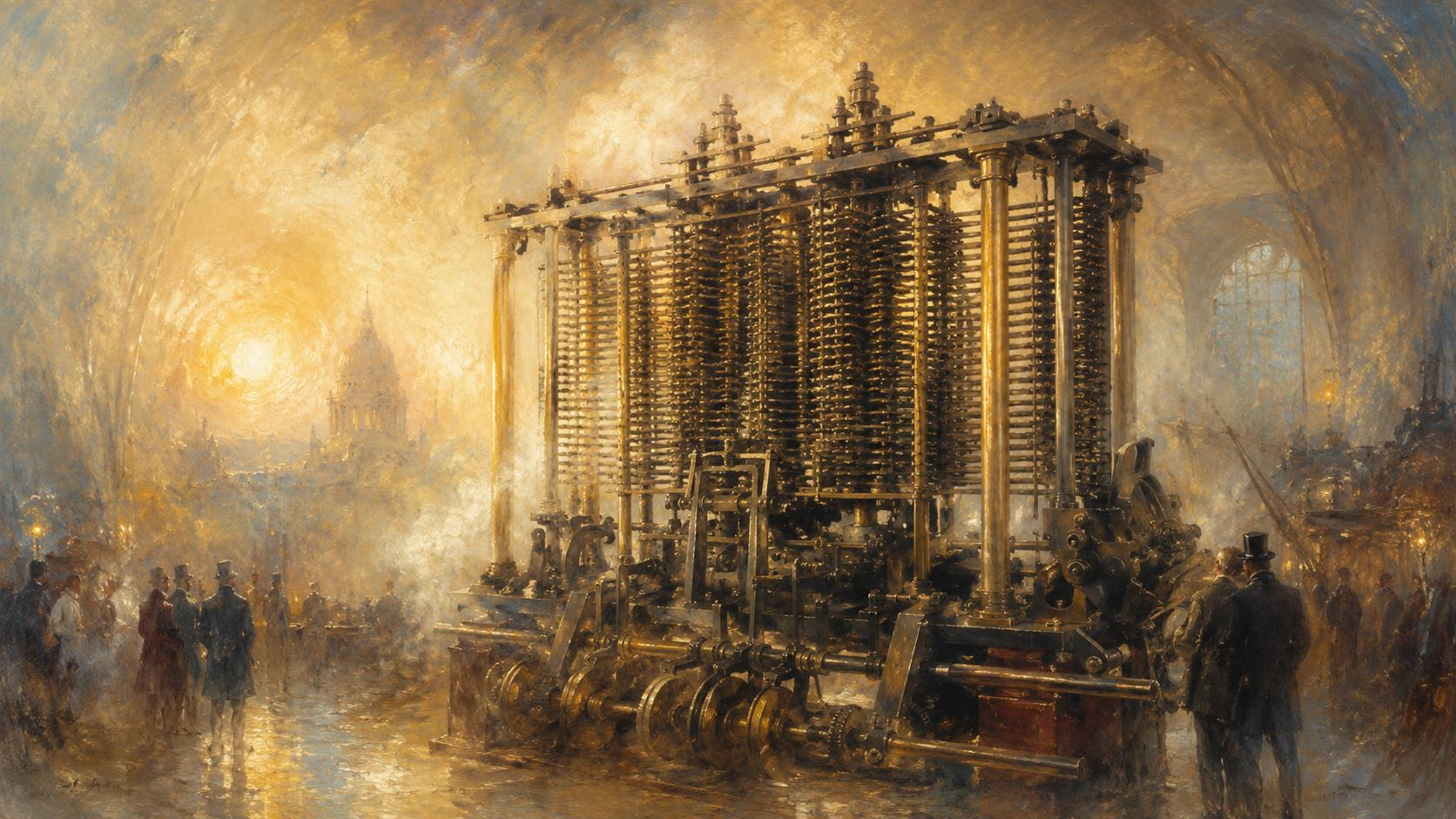
Humans and tools

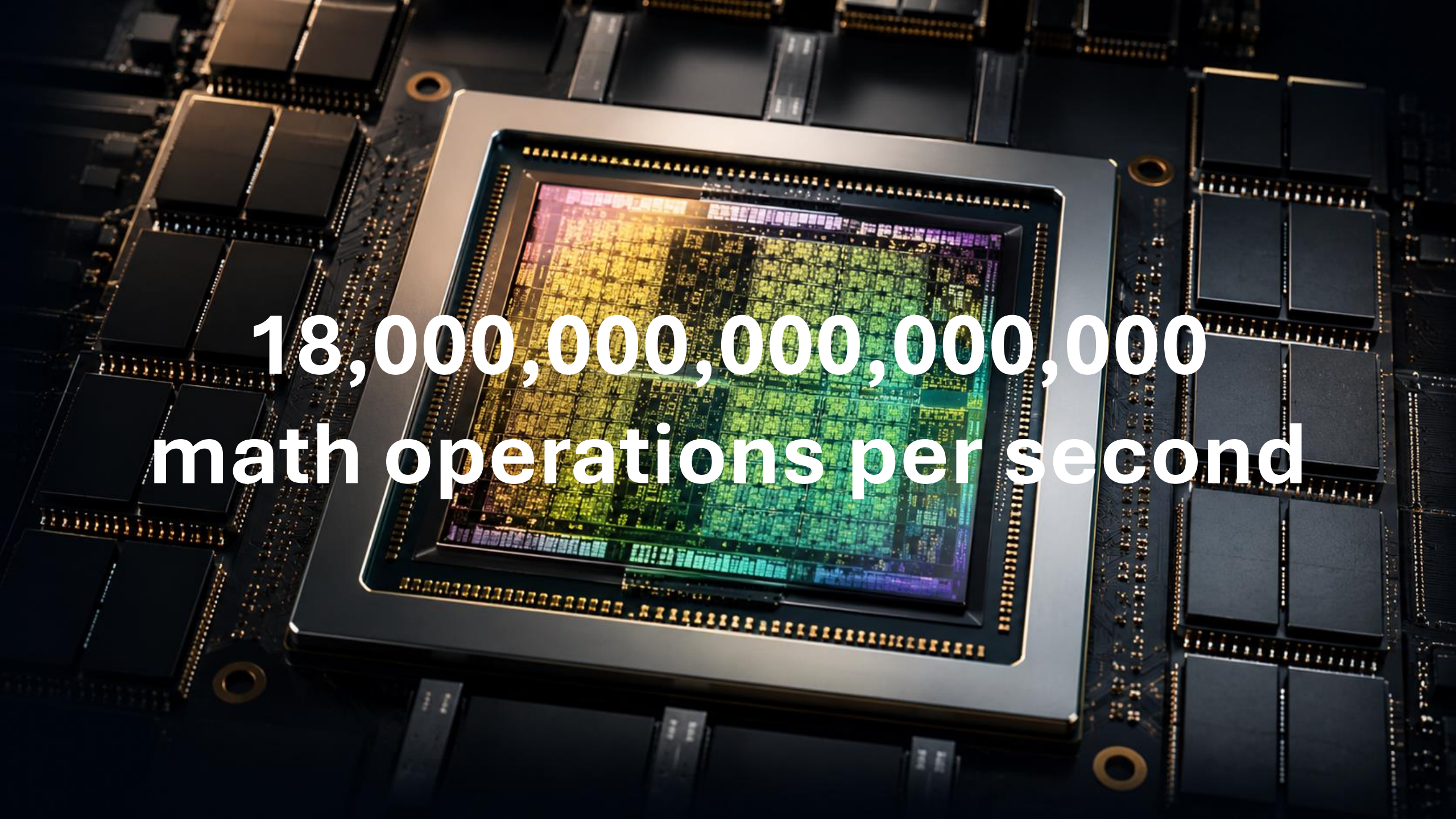
- Tools extended human abilities
 - Example: bashing one rock into another rock to get fruit out of a nut
- Tools replicated manual skills
 - Example: milling flour
- Tools following instructions
 - Example: Jacquard's loom
- Tools replicating menial mental work
 - Example: calculator, Babbage's Analytical Engine
- Tools replicating higher level mental work
 - Example: Generative AI







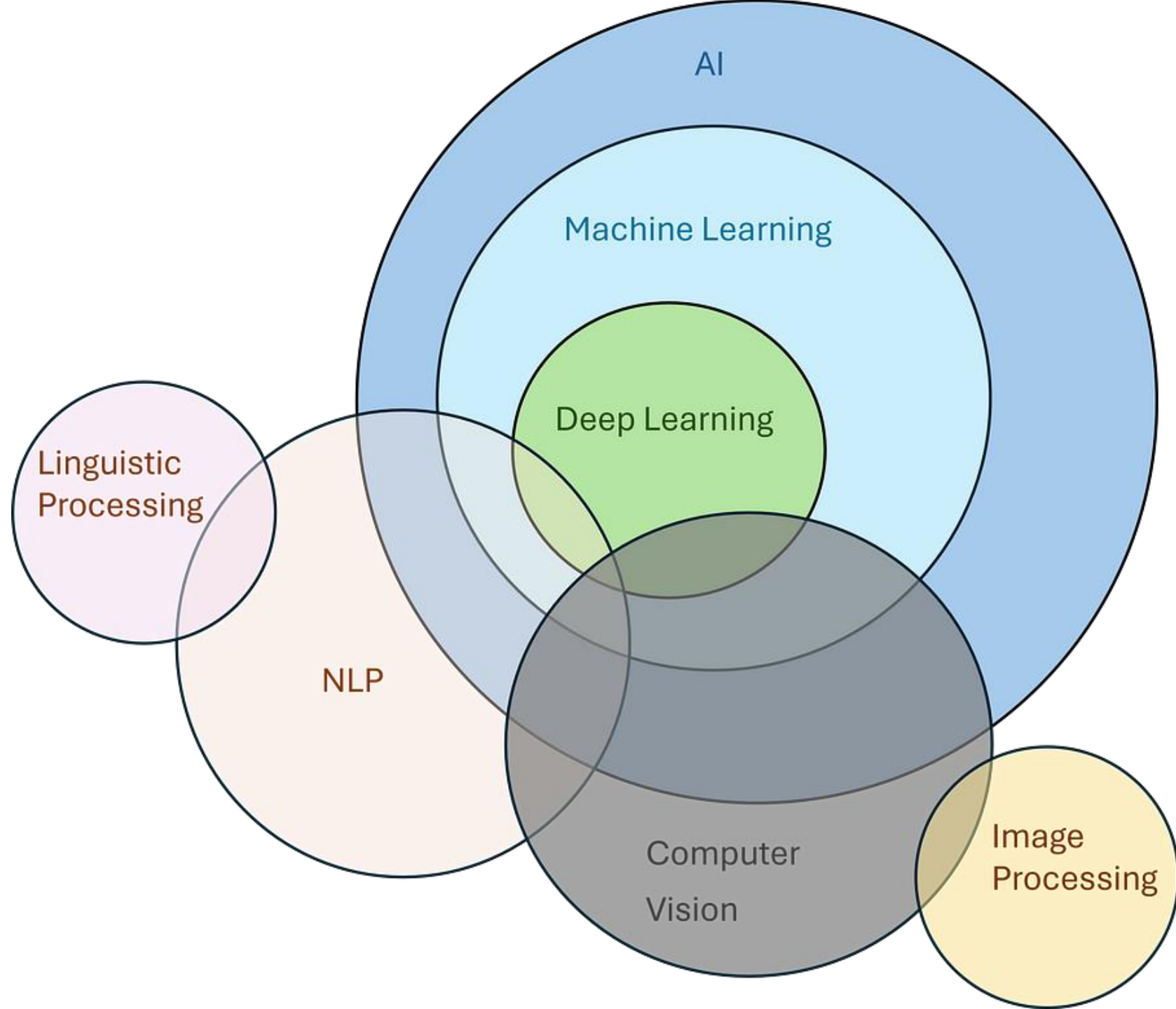




18,000,000,000,000,000
math operations per second

AI Definition

- computer systems capable of performing [or appearing to perform] tasks that typically require human intelligence, such as reasoning, learning, perception and language understanding.



How GenAI works

1. Tokenization

Breaking a sentence into tokens and mapping them to numbers.

The jury finds the defendant ____



Tokens

[The]

[jury]

[finds]

[the]

[defendant]

[____]



Token IDs
(numbers)

464

12874

5821

262

15449

?



Illustrative token IDs for presentation purposes.

2. Embedding

Converting tokens into dense vectors that capture patterns and meaning.

The jury finds the defendant _____



Tokens

[The]

[jury]

[finds]

[the]

[defendant]

[_____]



Embeddings
(dense vectors)

[0.18, -0.42,
0.77, 0.05]

[-0.31, 0.64,
0.22, -0.11]

[0.49, 0.13,
-0.28, 0.71]

[0.12, -0.37,
0.69, 0.02]

[-0.55, 0.81,
0.14, 0.33]

[?, ?, ?, ?]



Illustrative vectors for presentation purposes. Real embeddings use many more dimensions.

4,096

195632195340444710676146006269151967647891460822647583670440244955007979234806831653168063352410822127261412832649316026698524479144774820
635636689606323545307314411691666762162791394694092070291502323134941217187570147820695689607764989184145935498248425135187904518631201651
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746618960829801740285808978625080279572523536705887890228772737029147404532307451657142792826641145618188059651489067262642739432464434433
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357765264401098312772959159126520324927880722995575934629029990392746783529219852457853927553422398699220544171797664130504310896995378689
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718130815801619091928727351265900358931411000080829187986986740278404150069102819339866038299311965791248062834901160642021082211516254694
459374612340598933310732026532398080323125758353814136934304113106182019371433808396681230379824333016158922840050799189469921103676302277
300618938441111953929801531119785709778169871072386125842126166394582241026644294572656943338082326193960794118474170447394661625874864615
110308432945296740280854985701114364615906445302897850354710267093962063795293253880977553287000865984074914649811667970881246974684196929
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23426279185088654279550120978964678743218134089886700949155347482739679451709442521756299334127

3. Embedding Space

Token embeddings can be stored in a vector database, where related meanings cluster together.

The jury finds the defendant _____



Embedded
tokens

[The]

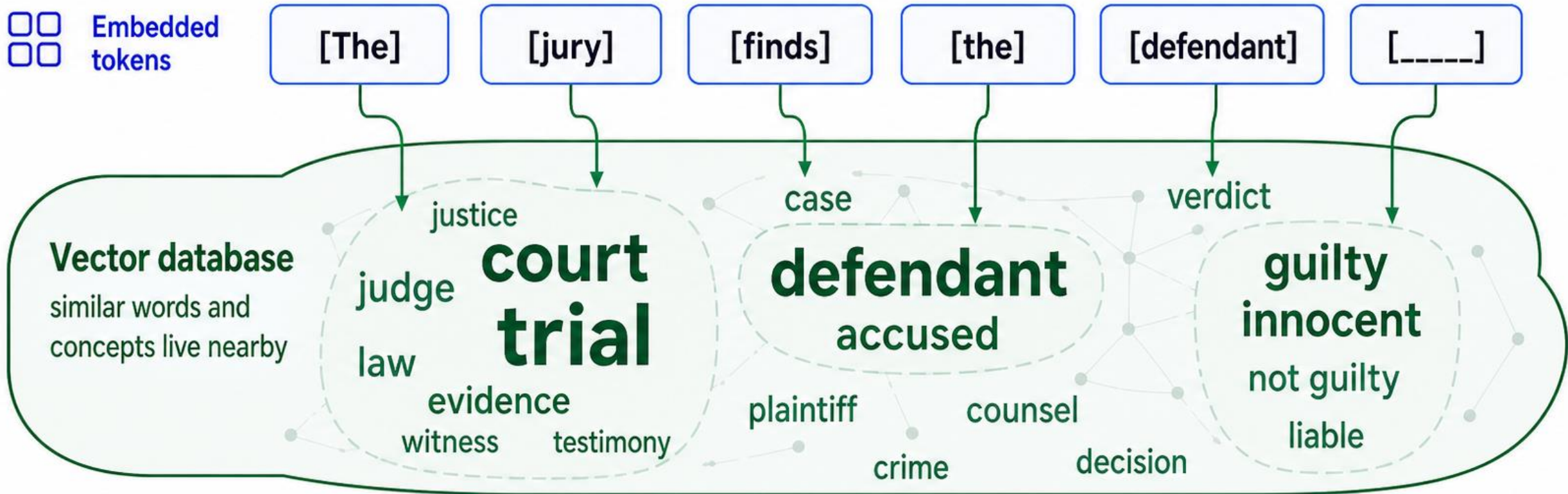
[jury]

[finds]

[the]

[defendant]

[_____]



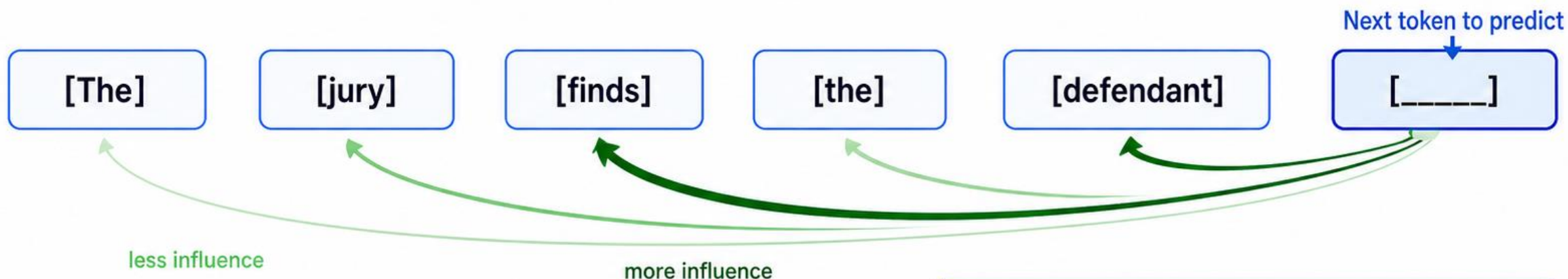
Illustrative word cloud only. Real vector databases store high-dimensional numerical embeddings, not words floating in space.

3.A

Attention

To predict the next token, the model weighs earlier tokens differently based on their relevance.

The jury finds the defendant ____



Attention weights from the prediction position

[The]	[jury]	[finds]	[the]	[defendant]
0.05	0.18	0.34	0.12	0.26

What attention is doing

- compares the current position to earlier tokens
- highlights the most useful context
- blends information before choosing the next token



Illustrative attention pattern only. Real models use many attention heads and layers, each tracking different relationships.

4. Temperature

Temperature changes how sharply the model favors the most likely next token.

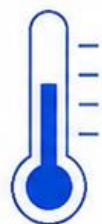
The jury finds the defendant __

Low temperature ($T = 0.2$)

Sharper distribution

guilty		72%
liable		14%
innocent		9%
negligent		5%

Sampled next token: [guilty]



Lower temperature
= more predictable

Higher temperature
= more varied

High temperature ($T = 1.2$)

Flatter distribution

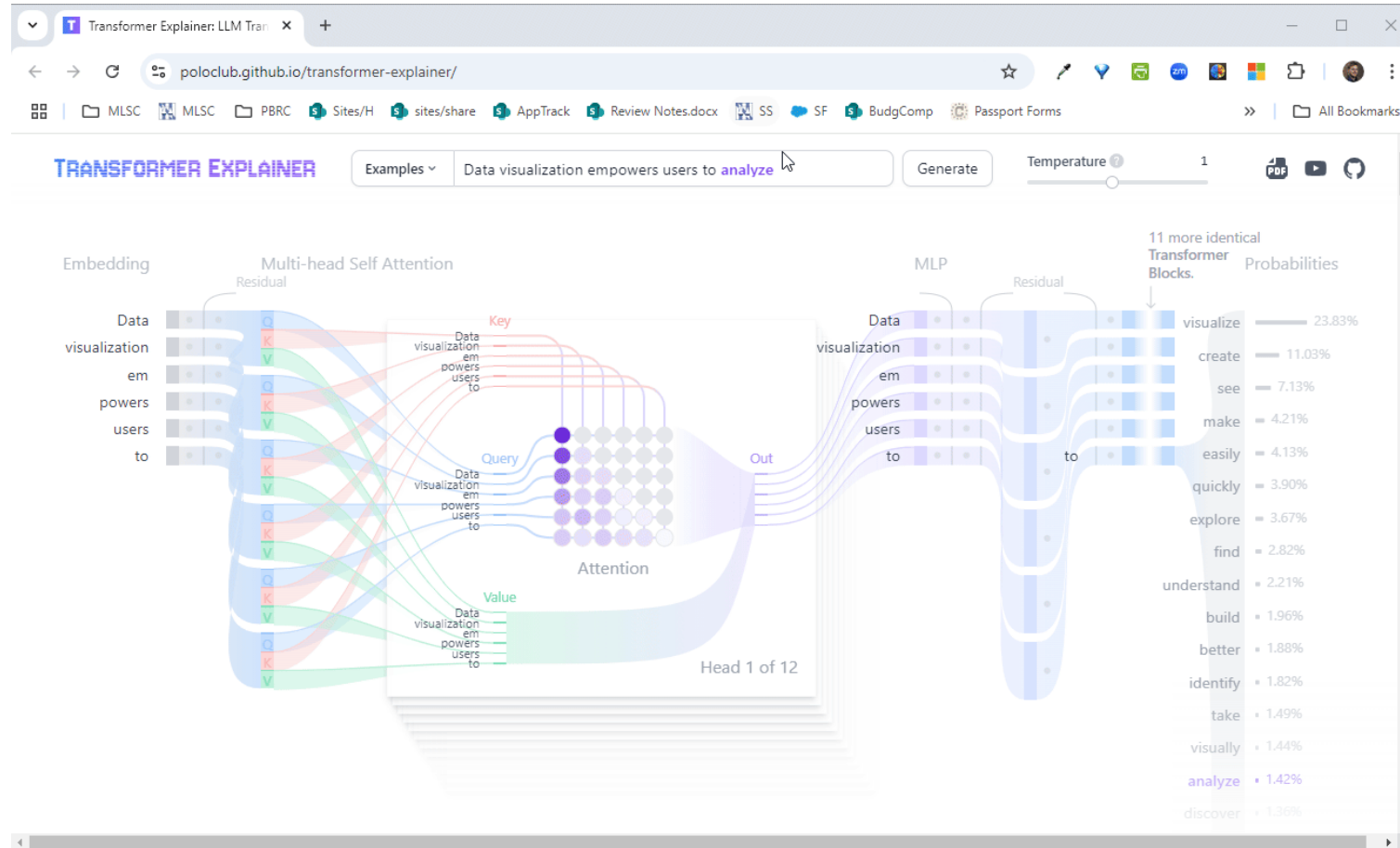
guilty		38%
liable		24%
innocent		20%
negligent		18%

Sampled next token: [liable]



Illustrative probabilities for presentation purposes. Temperature rescales the next-token distribution: lower values concentrate probability on top choices, while higher values spread probability across more options.

See how Gen AI works!



6.

Hallucination

A model can produce text that sounds credible even when key details are wrong or invented.

“another seminal case on racial segregation, Green v. Board of Education, 74 U.S. 483 (1954),...”



Model
output



Looks plausible: legal case name + citation format



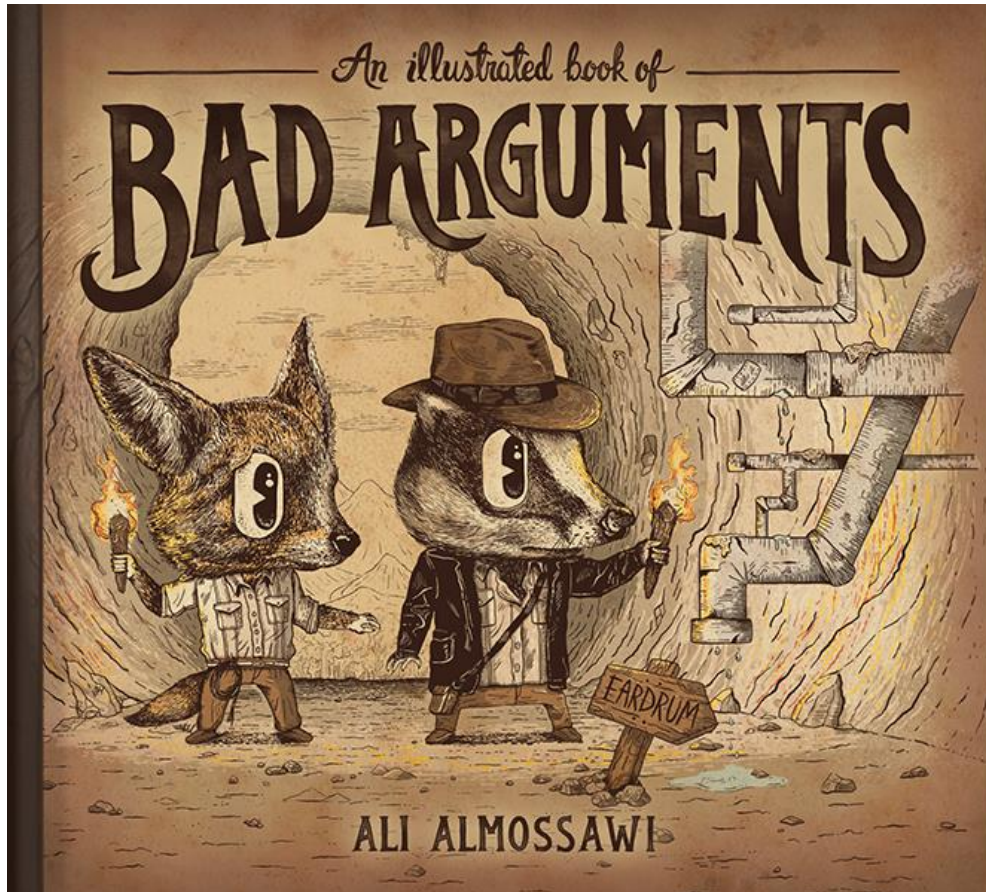
Problem 1: 'Green' is the wrong case name



Problem 2: '74' is the wrong reporter volume



Illustrative example only. Hallucinations are errors where a model generates fluent but incorrect information.



Examples

- [Case database](#) – 1,294 cases so far
- Research Hallucinations
 - Fabricated case or statute citations – most common
 - Fabricated quotes
 - Citing a case that does not stand for the proposition for which it's cited
- Drafting errors
 - Misstated facts
 - Inaccurate internal citations (case record or within document)

N.Z. v. Fenix International Limited

- Class action lawsuit against OnlyFans, fraud
- [Document 176-2](#): Declaration of attorney
- [Document 207](#): Order denying leave to amend briefs, imposing sanctions

N.Z. v. Fenix International Limited

ERROR CATEGORIES

The errors identified by defense counsel fall into the following categories:

1. A single hallucinated authority.
2. Authority cited for the correct proposition, but with an incorrect quotation or parenthetical.
3. Authority cited for the incorrect proposition and/or mischaracterized.
4. Incorrect record cites, including one resulting in mischaracterization of the Court's prior order.

#1 single hallucinated authority

The forum-selection clause in question is not enforceable against Plaintiffs asserting claims based on deception, and **public policy weighs against transfer where the gravamen of the action is fraud. See *Doe v. Match Group*, 2022 WL 4551660**, at 6 (C.D. Cal. Aug. 12, 2022). R.M. resides in

#1 single hallucinated authority

OnlyFans' Terms of Service. [CITE] [Did they make this argument before? Look at FNC briefing . . .] [. . . nature of the claims is relevant, right? Fraud/misrep . . .] R.M. resides in California, and the claims he asserts arise from deceptive conduct orchestrated by a California-facing platform. Venue is proper under 28 U.S.C. § 1391(b).

Given this language and the types of errors I have seen ChatGPT produce, I strongly suspect that at some stage in the drafting, ChatGPT extrapolated a proposition based on the note “[. . . nature of the claims is relevant, right? Fraud/misrep . . .],” and then fabricated a case to support the proposition.

#3 incorrect proposition

analytics cases like *Hulu*. [[CITE] . . . can we find a case on “partial” identifiers . . . ?].

Apparently in response to my “can we find a case” note, ChatGPT “filled in the blank” by fabricating a description of a case that met the criteria, and because it now had the *Calise* case in its memory, it just attributed the fabricated description to that case.

In short: the case entered via a draft placeholder; ChatGPT hallucinated the description of the case to match the placeholder; and the fabricated version was repeated in subsequent drafts. I failed to catch the error before filing, and it ended up in the filed brief.

Patterns of bad AI

- What the Science Says About Hallucinations in Legal Research

1. Models and data access
 1. Legal platforms generally have fewer hallucinations or inaccurate citations, but not enough to be reliable without checking
2. Sycophancy
3. Jurisdictional and geographic complexity
4. Knowledge cutoffs
5. Task complexity
6. Confidence paradox
7. Bonus: timing and effective date

Tool strategies for combating hallucinations

- Chain of thought reasoning
- Fine-tuning LLM with high quality human input
- Retrieval augmented generation
- Custom system prompts
- Architectural strategies
 - Orchestrator, mixture of experts

Chain-of-Thought Prompting

Step-by-step prompts can guide the model to break a task into smaller parts and reuse earlier outputs as additional context.

Prompt: Complete the sentence and explain your answer
The jury finds the defendant ____

1. Parse the task

- predict the next token
- explain the choice
- note the legal context

2. Generate candidates

Possible next tokens:

guilty

liable

innocent

negligent

3. Compare and narrow

criminal verdict → guilty / innocent

civil finding → liable / negligent

best fit depends on context

4. Produce the response

Next token: [guilty]

Short explanation: the phrase strongly suggests a verdict.



Growing context

Each step adds outputs that the next step can use.

Card A:
Original prompt

The jury finds the defendant ____.

Card B:
Step 1 output

Task =
predict + explain

Card C:
Step 2 output

Candidates =
guilty, liable,
innocent,
negligent.

Card D:
Step 2 output

Most likely in
criminal context =
guilty

Card E:
Final response

The jury finds the
defendant **guilty**,
because it matches the
courtroom-verdict pattern.

Why this helps

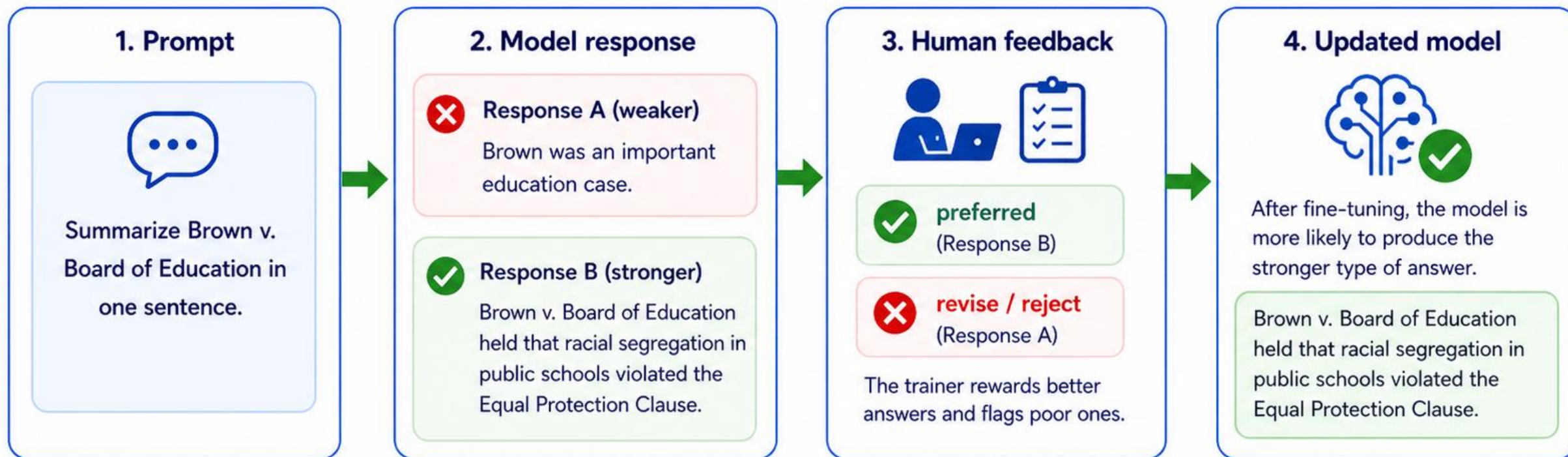
- complex tasks become smaller tasks
- earlier outputs expand the working context
- the answer becomes more coherent and consistent



Illustrative workflow only, Real models do not literally store thoughts as plain-English notes, but step-by-step prompting can improve decomposition and keep more useful context active during generation.

Fine-Tuning

Human feedback helps adjust the model so future responses are more aligned, accurate, and useful.



What changes



better tone and clarity



more consistent answers



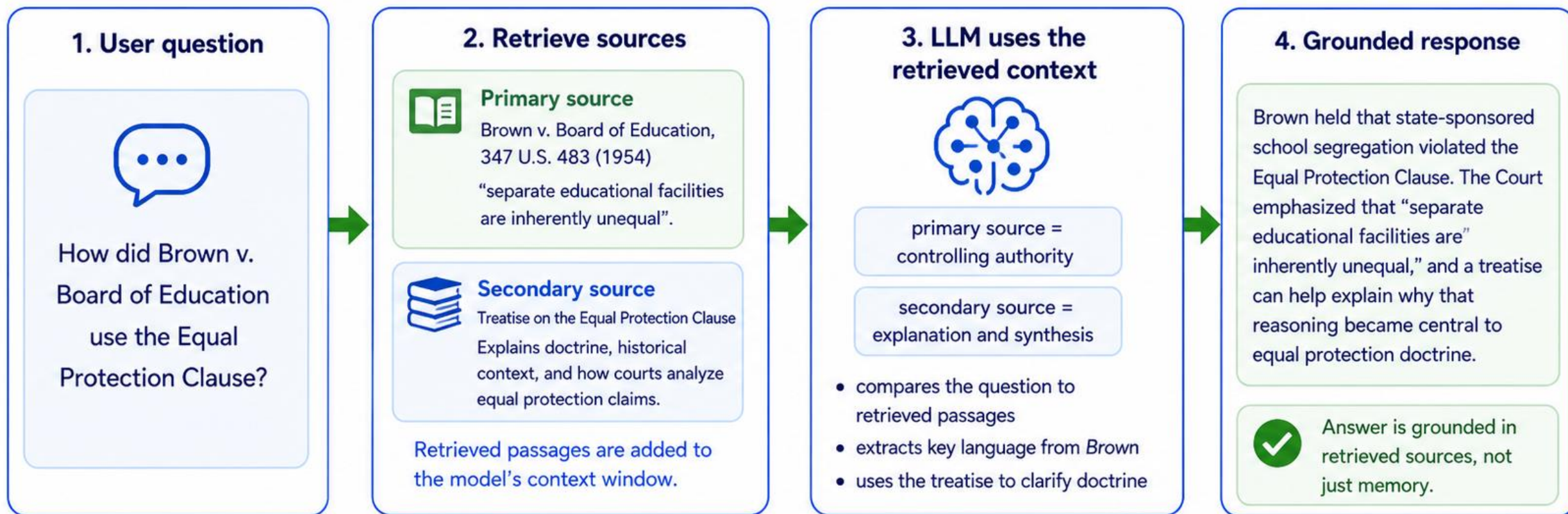
closer alignment with human preferences



Illustrative example only. Fine-tuning adjusts the model based on training examples and human judgment; it does not make the model perfect.

Retrieval-Augmented Generation

The model retrieves outside sources, adds them to context, and then generates a more grounded answer.



Why RAG helps



more accurate answers



uses current or authoritative sources



can cite or summarize supporting materials



Illustrative example only. RAG improves grounding by supplying source material, but the model can still misread or misstate sources.

System prompt

- A system prompt is an instruction set that defines the LLMs responses
- [Claude's system prompt](#) for the web and mobile apps
- [OpenAI's system prompt for Codex](#)
- Can be customized in some ways by user preference

Can fundamental paradox be overcome?

Legal research is a theory-of-knowledge problem: GenAI can help at the semantic level by defining concepts, identifying associations, and generating summaries. Legal knowledge requires epistemic justification through authoritative sources, and many legal tasks still require true symbolic reasoning, analogical validation, and human judgment.